



Iranian Mathematical Society Competition

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by mojyla222

– Day 1

P1 Let $a, b, c, d > 0$. Prove that for every positive real number x , the sequence $\{f^{2n}(x)\}_{n=1}^{\infty}$ converges, where $f(x) = \frac{ax+b}{cx+d}$. (Here, f^n denotes the n -th iterate of f .)

P2 Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two continuous functions such that $f \circ g$ is bijective. Prove that both functions f and g are bijective. (A bijective function is one that is both injective and surjective.)

P3 Let R be a ring with unity such that $2 \in R$ is not a zero divisor. Let $x \in R$ such that x commutes with every invertible element of R . Prove that every idempotent element in R also commutes with x .

P4 Let (X, d) be a metric space. For every $x \in X$ and $r > 0$, define:

$$B_r(x) = \{y \in X \mid d(x, y) < r\} \quad \text{and} \quad C_r(x) = \{y \in X \mid d(x, y) \leq r\}.$$

Prove that if X is compact and x_0 is a point in X , then for at most countably many values of r , we have

$$\overline{B_r(x_0)} \neq C_r(x_0).$$

P5 Let X be a set and $T : X \rightarrow X$ a function such that for every subset $A \subseteq X$, the set $A \setminus T(A)$ is countable. Prove that the set $\{x \in X \mid T(x) \neq x\}$ is countable.

P6 Let G be a group such that for every $a, b \in G$, we have $\langle a \rangle \cap \langle bab^{-1} \rangle = \{e\}$ or $\langle a \rangle \cap \langle bab^{-1} \rangle = \langle a \rangle$. Also assume that $x, y \in G$ are non-trivial elements such that $xy = yx$ and $(o(x), o(y)) = 1$. If $\langle x \rangle$ is a normal subgroup in G then prove that $\langle y \rangle$ is also a normal subgroup of G .

– Day 2

P7 Give an example of a metric space that has at most countably many compact subsets but uncountably many closed subsets.

P8 Let S be a semigroup with an identity element. Suppose x is an element of S such that x^2 is invertible. Prove that x is also invertible.

P9 Prove that the number $2^{1398} + 1$ has at least three distinct prime factors.

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- P10** Let F be a field of characteristic zero, and let $A, B \in M_n(F)$. Prove that if every linear combination of A and B is nilpotent, then for every $1 \leq k$, we have $\text{Tr}(A^k B) = 0$.
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- P11** Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an analytic function. For $r > 0$, let $S_r = \{z \in \mathbb{C} \mid |z| = r\}$. Suppose that $\text{diam} f(S_1) \leq 1$. Prove that for all $0 < r < 1$ $\text{diam}(f(S_r)) \leq r$.
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- P12** Consider two disjoint families $\mathcal{F}_1$ and $\mathcal{F}_2$ of subsets of $\{1, 2, \dots, n\}$ such that for each $i \in \{1, 2\}$ and every $A, B \in \mathcal{F}_i$, we have $A \cap B \neq \emptyset$. Show that the set $\{1, 2, \dots, n\}$ can be colored with three colors (white, red, blue) such that every member of $\mathcal{F}_1 \cup \mathcal{F}_2$ with at least three members has at least two colors.
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